

Significance Level (α) – Concept & Examples

1. What is Significance Level (α)?

The significance level (α) is the probability of rejecting a true null hypothesis (Type I error) in hypothesis testing. It represents the level of risk we are willing to take when making conclusions about a population based on sample data.

Common Significance Levels

- 0.10 (10%) → Less strict (higher chance of Type I error)
- 0.05 (5%) → Most common
- 0.01 (1%) → More strict (lower chance of Type I error)

If $\alpha = 0.05$, it means that there is a 5% probability of rejecting a true null hypothesis.

2. Relationship Between Significance Level & Confidence Level

The significance level α is complementary to the confidence level:

$$\text{Confidence Level} = 1 - \alpha$$

Significance Level (α)	Confidence Level ($1 - \alpha$)
0.10 (10%)	90% Confidence
0.05 (5%)	95% Confidence
0.01 (1%)	99% Confidence

For example, if we choose $\alpha = 0.05$, we are 95% confident that we are making a correct decision.

3. Significance Level in Hypothesis Testing

Steps in Hypothesis Testing

- State the Hypotheses:
 - Null Hypothesis (H_0): No effect or no difference.

- Alternative Hypothesis (H_1): There is an effect or difference.
2. **Choose a Significance Level (α):**
 - Typically 0.05 (5%) is used.
 3. **Calculate the Test Statistic:**
 - Compare the sample result to the hypothesized population value using a test statistic (Z, t, F, or χ^2).
 4. **Find the p-value:**
 - The p-value is the probability of obtaining a result **at least as extreme** as the one observed, assuming H_0 is true.
 5. **Compare p-value with α :**
 - If p-value $\leq \alpha$, reject H_0 (statistically significant).
 - If p-value $> \alpha$, fail to reject H_0 (not statistically significant).
-

4. Numerical Examples

Example 1: Hypothesis Testing with $\alpha = 0.05$

Problem:

A company claims that the average lifespan of their light bulbs is **1000 hours**. A sample of $n = 40$ bulbs has an average lifespan of **980 hours** with a standard deviation of **50 hours**. Test if the claim is true at $\alpha = 0.05$.

Step 1: State the Hypotheses

- $H_0: \mu = 1000$ (The mean lifespan is 1000 hours)
- $H_1: \mu \neq 1000$ (The mean lifespan is different from 1000 hours)

Step 2: Select $\alpha = 0.05$

Since it is a **two-tailed test**, the critical value for Z at $\alpha = 0.05$ is ± 1.96 .

Step 3: Compute the Test Statistic

Using the Z-test formula:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$
$$Z = \frac{980 - 1000}{50 / \sqrt{40}}$$
$$Z = \frac{-20}{7.91} = -2.53$$

Step 4: Compare with Critical Value

- $Z = -2.53$
- Critical Value = ± 1.96 (from Z-table)
- Since $-2.53 < -1.96$, we reject H_0 .

Conclusion:

At $\alpha = 0.05$, we reject the null hypothesis and conclude that the light bulbs’ average lifespan is significantly different from 1000 hours.

Example 2: p-value Approach

In the previous problem, we found $Z = -2.53$.
Using the Z-table, the p-value for $Z = -2.53$ is 0.0057.
Since p-value (0.0057) < α (0.05), we reject H_0 .
Thus, the result is statistically significant.

5. Type I & Type II Errors

Decision	H_0 is True	H_0 is False
Reject H_0	Type I Error (False Positive)	Correct Decision
Fail to Reject H_0	Correct Decision	Type II Error (False Negative)

- Type I Error (False Positive): Rejecting H_0 when it is actually true (controlled by α).
- Type II Error (False Negative): Failing to reject H_0 when it is actually false (controlled by sample size & power).

Lowering α reduces Type I Error but increases Type II Error.

6. Choosing the Right α

α Level	Usage
0.10 (10%)	When consequences of a Type I error are low (e.g., market research).
0.05 (5%)	Standard choice in most research studies.
0.01 (1%)	When consequences of a Type I error are high (e.g., medical trials, fraud detection).

For example:

- **A new drug test** → Use $\alpha = 0.01$ (minimize false positives).
 - **Website A/B testing** → Use $\alpha = 0.05$ (balance between errors).
 - **Exploratory studies** → Use $\alpha = 0.10$ (less strict).
-

7. Summary

1. **Significance Level (α)** represents the probability of making a **Type I error**.
2. Common values: **0.10, 0.05, 0.01**.
3. **Confidence Level** = $1 - \alpha$.
4. If **p-value** $\leq \alpha$, we **reject H_0** .
5. Lower α reduces **Type I error** but increases **Type II error**.
6. The choice of α depends on the **study's risk and consequences**.